

GeoPT: Privileged Geometric Post-Training for One-Step Whole-Body Motion Planning in Mobile Manipulation

Project page: <https://geopt-commits.github.io/>

Abstract—Flow and diffusion models have recently emerged as effective generative approaches for motion planning in mobile manipulation. However, three challenges remain: collecting demonstrations that cover diverse whole-body interactions is difficult; their iterative sampling procedures can make local replanning computationally expensive; and imitation-trained planners often degrade when deployed beyond the support of their training data. To address these challenges, we present GeoPT, a privileged geometric post-training framework for one-step whole-body motion planning. An automated simulator constructs a synchronized data pipeline of expert plans, robot states, and egocentric range measurements to train a diffusion-based behavior prior. This prior initializes a compact one-step Gaussian local planner. During post-training, predicted action chunks are flagged for geometric correction when scene or self-collision occurs. For these collision cases, differentiable collision distances correct unsafe plans, while a frozen DDPM behavior prior constrains policy drift and preserves pretrained task behavior. Privileged geometry is used only during training. We evaluate GeoPT in obstacle avoidance, goal-reaching, and pick-and-place tasks, including real-robot deployment, demonstrating robust whole-body motion planning with efficient inference.

I. INTRODUCTION

Flow and diffusion models effectively parameterize motion planners by representing temporally coherent action sequences and multimodal behaviors [1], [2]. They have recently been applied to whole-body trajectory generation for mobile manipulation [3]. Local planning in cluttered environments, however, remains constrained by three factors. Demonstrations must cover diverse coordinated base–arm interactions, iterative generative sampling can be too slow for frequent replanning, and imitation-trained planners may deteriorate beyond the support of the demonstrations.

Existing mobile-manipulation datasets [4] consist of trajectories connecting specified start and goal states in known scenes, without the egocentric observations needed for real-time local planning. We extend this setup with an automated simulation pipeline that runs an expert whole-body planner [5] while synchronously recording robot states, target poses, and egocentric range measurements along long-horizon tasks. The synchronized records are segmented into action chunks with a fixed chunk size, while the range measurements are encoded into a height-stratified polar occupancy representation that preserves obstacle structure across heights with low processing overhead.

Diffusion planners can be inefficient for real-time deployment, while planners trained only on expert demonstrations may degrade on states encountered during execution. Iterative denoising requires multiple network evaluations, incurring high replanning latency. Small prediction errors



Fig. 1. Real-world deployment of GeoPT.

alter the base–arm configuration, expose new local geometry, and change the next planning input.

To expose and correct these failure states, we present **GeoPT**, a privileged geometric post-training framework for one-step whole-body motion planning. A conditional diffusion provides an expressive behavior prior from which we initialize a compact Gaussian local planner. The planner is rolled out using only deployable observations, and states whose predicted action chunks trigger scene or self-collision are retained. Differentiable collision distances correct the predicted plans, while a frozen DDPM behavior prior constrains policy drift to preserve pretrained task behavior. Privileged geometry is used only during post-training, and the planner produces an action chunk in one forward pass.

This design separates two roles in motion planning: the behavior prior preserves useful task structure, whereas collision-triggered post-training corrects execution failures. GeoPT couples failure-state collection with privileged geometric supervision while retaining low-latency deployment.

The key contributions of this letter are:

- **Synchronized expert-data pipeline:** We collect synchronized expert motion plans, robot states, target poses, and egocentric range observations for whole-body mobile manipulation, providing action-chunk samples for behavior-prior training and local-planner evaluation.

- **Effective collision-triggered post-training:** GeoPT collects collision-triggered states and applies privileged differentiable geometric supervision to improve safety without requiring expert labels for these states.
- **Simulation and real-world validation:** We evaluate the complete framework in simulation and deploy it on a real mobile manipulator, demonstrating reduced collision rates and low-latency whole-body motion planning.

II. RELATED WORK

A. Whole-Body Planning and Expert Motion Generation

Sampling-based search and trajectory optimization are foundational tools for high-dimensional motion planning and collision avoidance [6], [7]. Whole-body mobile-manipulation systems extend these tools through environment-adaptive search and feasibility-aware optimization [5], [8], while hybrid methods combine learned policies with model-based control [9]. Map-based diffusion planners further provide expressive whole-body trajectories [3]. These methods are effective sources of expert supervision, but typically generate trajectories from an explicit scene model and a specified task endpoint. They therefore do not directly provide the synchronized egocentric observations needed for local replanning. GeoPT uses an expert planner offline to generate trajectory–perception records, while the deployed policy replans from egocentric range measurements after each action chunk.

B. Generative Policies and Diffusion-Based RL

Flow and diffusion policies model expressive robot behaviors and temporally coherent action sequences [1], [2]. Progressive and consistency distillation reduce the sampling cost of diffusion models [10], [11], while OneDP directly distills a diffusion policy for one-step visuomotor control [12]. These methods primarily improve policy expressivity or approximate iterative diffusion sampling; they do not specifically correct failures encountered during deployment. Beyond imitation learning, diffusion behavior models have also been used as priors or regularizers for offline reinforcement learning [13], [14]. The implicit action distribution of diffusion policies makes direct RL optimization difficult. GeoPT avoids this issue by using a Gaussian surrogate policy for geometric post-training, while retaining the diffusion model as a behavior prior. Related mobile-manipulation policies address unseen or dynamic environments through learned navigation and control [15], but do not use collision-triggered geometric post-training in the same way.

C. Post-Training for Policy Adaptation

Post-training methods refine an initially learned policy using additional states, costs, or privileged information. DAgger addresses distribution shift by querying an expert on learner-visited states [16]. Privileged guidance and differentiable geometric objectives provide safety supervision beyond expert demonstrations [17], [18]. These methods motivate adapting a policy on states that are not well represented in the original data, but may require online expert queries

or retain task-specific guidance during execution. GeoPT instead performs collision-triggered post-training: it collects states from policy rollouts, applies privileged scene and self-collision distances to those states, and discards the geometric checker at deployment. A frozen diffusion behavior prior regularizes the Gaussian policy so that geometric correction improves safety without sacrificing the pretrained behavior.

III. METHODOLOGY

GeoPT is a privileged geometric post-training framework for one-step whole-body motion planning in mobile manipulation. The method has four stages: (i) automated trajectory and perception data generation, (ii) behavior-prior initialization with a DDPM and a one-step Gaussian local planner, (iii) collision-triggered learner-state collection, and (iv) privileged geometric post-training. Privileged geometry is used only during training; the deployed planner receives the same observation modalities as a robot operating online.

A. Problem Formulation

We consider real-time whole-body motion planning for a mobile manipulator from local observations. At time t , let $s_t = [x_t, y_t, \psi_t, q_{1,t}, \dots, q_{6,t}] \in \mathbb{R}^9$ denote the current base pose and arm joint positions, and let $g_t \in \mathbb{R}^9$ denote the target pose. The planner receives

$$o_t = \{r_t, s_t, g_t\}, \quad (1)$$

where r_t is a bounded egocentric range observation, and predicts an action chunk

$$\begin{aligned} \mathbf{A}_t &= [a_t, \dots, a_{t+H-1}], \\ a_\tau &= [x_\tau, y_\tau, \psi_\tau, q_{1,\tau}, \dots, q_{6,\tau}] \in \mathbb{R}^9. \end{aligned} \quad (2)$$

which is executed by a low-level controller before the next observation. Our objective is to learn a local planning policy $\pi_\theta(\mathbf{A}_t \mid o_t)$ that repeatedly generates goal-directed, collision-free action chunks with low replanning latency. Scene geometry and a full robot collision model are privileged information available only for training; deployment uses o_t alone.

B. Expert Data Generation

An automated data-generation pipeline couples a map-based expert planner [5] with a first-person sensor simulator [19] that provides raw LiDAR point clouds. At each local replanning instant, it records the expert trajectory, robot state, target pose, and point cloud. Sliding windows over these records form the initial dataset $\mathcal{D}_0 = \{(o_t, \mathbf{A}_t^*)\}$. We use $H = 50$ and a temporal interval of $\Delta t = 0.03$ s between consecutive states. This produces aligned state–action samples rather than only complete trajectories, allowing initialization and post-training rollouts to use the same local-policy task. Across 30 scenes, we collect 100 trajectories per scene for obstacle avoidance, goal reaching, and pick-and-place. Slicing yields approximately 150,000 training samples.

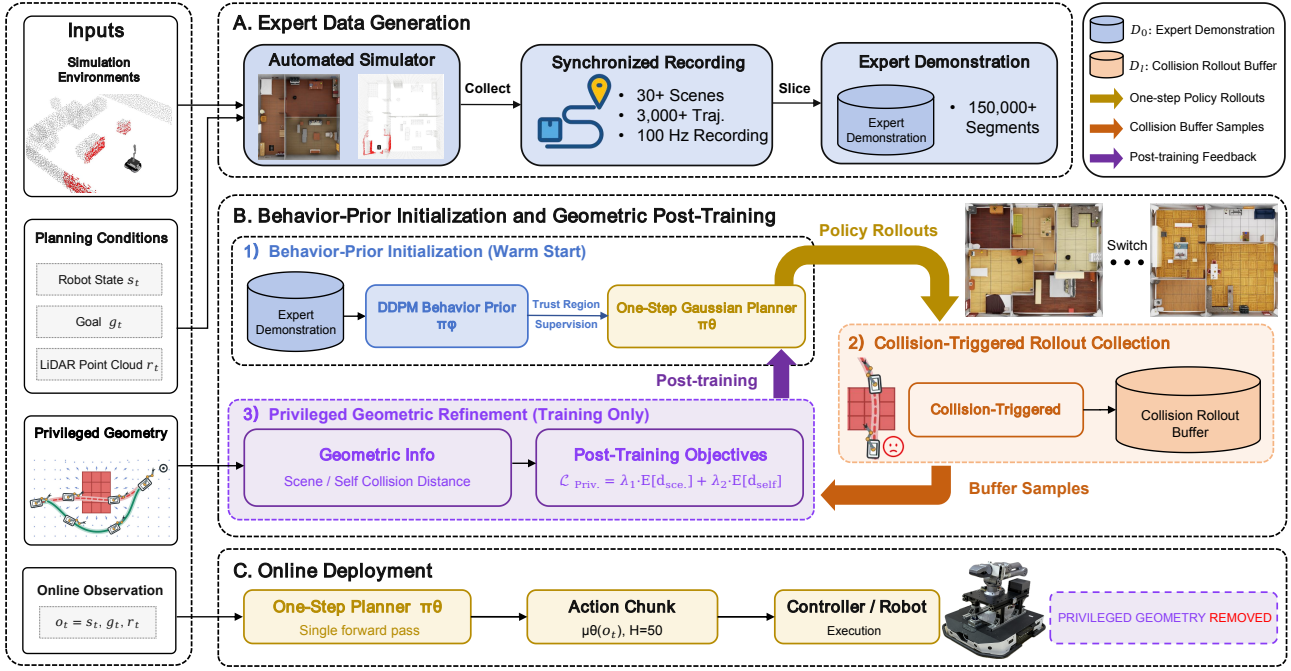


Fig. 2. **Overview of GeoPT.** Expert rollouts are collected in an automated simulator and synchronized into the fixed demonstration buffer $\mathcal{D}_0$. The initialized one-step planner is then rolled out with deployable observations; scene and self-collision checks trigger collection of failure states into $\mathcal{D}_{\text{learner}}$. A 1:1 mixture of the original and learner buffers drives geometric correction together with frozen DDPM behavior regularization. Privileged geometry is used only in this training loop and is removed at deployment, where the planner produces an action chunk from the observation, robot state, and goal.

C. Behavior-Prior Initialization

Multimodal condition encoder. The egocentric observation r_t captures the local obstacle geometry around the robot. It is obtained by converting the raw LiDAR point cloud collected in Sec. III-B into a height-stratified polar range image. The interval $[0.1, 1.6]$ m is divided into 30 slices of width 0.05 m, and each slice is sampled by 180 rays at a 2° angular resolution. Each bin stores the nearest return, clipped at 5 m and normalized by this range. The resulting raycasting representation is illustrated in Fig. 3. A circular 1-D CNN encodes the range image along the angular dimension and produces a sequence of environment tokens $\mathbf{z}_t^r$. The robot state s_t , which contains the current base pose and arm joint positions, and the goal g_t , which specifies the target base pose and arm configuration, are separately mapped by MLP encoders to the state token $\mathbf{z}_t^s$ and goal token $\mathbf{z}_t^g$. The three modalities are concatenated and fused by residual multi-head self-attention:

$$\begin{aligned} \mathbf{z}_t^r &= E_r(r_t), \\ \mathbf{z}_t^s &= E_s(s_t), \\ \mathbf{z}_t^g &= E_g(g_t), \\ c_t &= E(o_t) = F_{\text{attn}}([\mathbf{z}_t^r; \mathbf{z}_t^s; \mathbf{z}_t^g]). \end{aligned} \quad (3)$$

where E_r denotes the circular CNN, E_s and E_g denote the state and goal MLPs, and F_{attn} denotes residual multi-head self-attention. The complete multimodal condition encoder is denoted by $E(\cdot)$, and its output $c_t = E(o_t)$ is the fused condition token sequence.

DDPM behavior prior. The diffusion noise-prediction module is a DiT that cross-attends to c_t . We denote the complete conditional DDPM behavior prior by π_ϕ ; its reverse

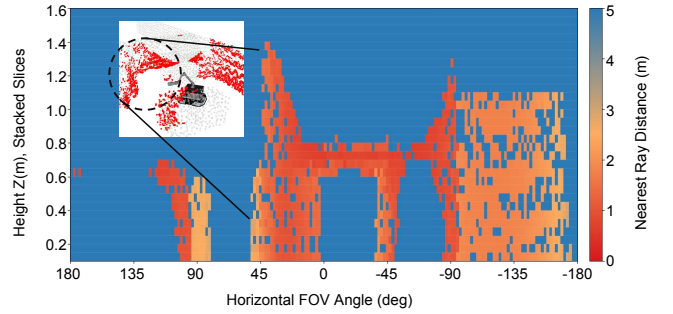


Fig. 3. **Height-stratified polar raycasting observation.** The LiDAR point cloud is partitioned into vertical slices and sampled along uniformly spaced angular rays to form the range representation used by the multimodal condition encoder.

diffusion process is parameterized by the noise-prediction network ϵ_ϕ , which defines the action distribution $\pi_\phi(\mathbf{A}_0 | o)$. We train this prior on $\mathcal{D}_0$. Let p_K denote the training distribution over diffusion indices $k \in \{1, \dots, K\}$, where $K = 50$ is the total number of denoising steps. Given a clean action chunk $\mathbf{A}_0$ and $k \sim p_K$, the forward process is

$$\mathbf{A}_k = \sqrt{\bar{\alpha}_k} \mathbf{A}_0 + \sqrt{1 - \bar{\alpha}_k} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (4)$$

and the DDPM behavior prior is optimized with the standard noise-prediction objective

$$\mathcal{L}_{\text{DDPM}}(\phi) = \mathbb{E}_{(o, \mathbf{A}_0) \sim \mathcal{D}_0, k \sim p_K, \epsilon} \left[\|\epsilon - \epsilon_\phi(\mathbf{A}_k, k, c_t)\|_2^2 \right]. \quad (5)$$

During training, $E(\cdot)$ and the DiT are jointly trained from scratch on $\mathcal{D}_0$. The pretrained weights of $E(\cdot)$ are then transferred to the Gaussian policy and fine-tuned jointly with its policy head. This transfer gives the one-step policy

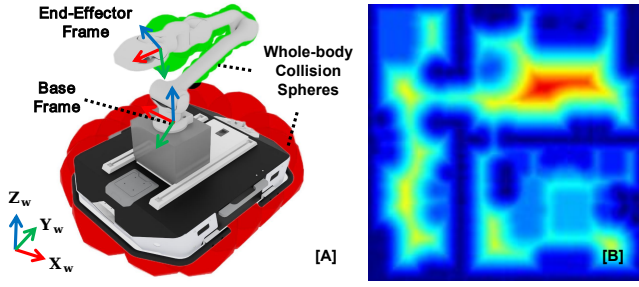


Fig. 4. **Privileged collision representation used during post-training.** (a) Whole-body collision geometry of the mobile manipulator. (b) Scene-distance representation used by the differentiable collision checker.

a behavior-aware warm start with task-relevant multimodal representations inherited from the expert data.

One-step Gaussian policy. Using the transferred encoder, the Gaussian policy obtains $c_t = E(o_t)$. A residual MLP block maps c_t to two linear heads that predict the action-chunk mean and diagonal log-standard deviation:

$$\pi_\theta(\mathbf{A}_t | o_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(c_t), \text{diag } \boldsymbol{\sigma}_\theta^2(c_t)). \quad (6)$$

Deployment uses $\mathbf{u}_t = \boldsymbol{\mu}_\theta(c_t)$ as the deterministic output.

In high-dimensional and sparsely sampled observation spaces, action diversity can arise from two sources: multiple feasible continuations at a given observation and rapid changes in locally effective actions across nearby observations [20]. For a fixed observation, let $p(\mathbf{A} | o)$ denote the conditional action distribution induced by the data. Action MSE draws a unimodal Gaussian policy toward the arithmetic mean of distinct continuations, which can produce an infeasible action. To retain this distributional structure during one-step initialization, we evaluate Gaussian-policy actions against the frozen DDPM behavior prior. For a Gaussian-policy sample $\mathbf{A}_\theta \sim \pi_\theta(\cdot | o)$, we apply the DDPM forward process using $k \sim p_K$:

$$\tilde{\mathbf{A}}_k = \sqrt{\bar{\alpha}_k} \mathbf{A}_\theta + \sqrt{1 - \bar{\alpha}_k} \epsilon. \quad (7)$$

The frozen DDPM behavior prior predicts the noise component, and the behavior-preservation loss is

$$\mathcal{L}_{\text{TR}}(\theta) = \mathbb{E}_{o, k \sim p_K, \epsilon, \mathbf{A}_\theta} \left[\left\| \epsilon - \epsilon_\phi(\tilde{\mathbf{A}}_k, k, c_t) \right\|_2^2 \right]. \quad (8)$$

Under the denoising approximation, a low value of Eq. 8 indicates that the Gaussian-policy action lies in a high-density region of the behavior-prior distribution. The objective therefore guides the Gaussian policy toward a locally effective continuation supported by the behavior prior. During privileged post-training, the same loss serves as a behavior-preservation term that penalizes policy updates departing from the pre-trained behavior.

D. Collision-Triggered Privileged Post-Training

GeoPT repeatedly rolls out the current policy to identify collision failures and then corrects these failures using privileged geometric supervision. The complete rollout-update loop is summarized in Algorithm 1. Policy rollouts use the deployment observation o_t , while a frozen behavior prior and

a differentiable collision model provide complementary training signals. This separation concentrates geometric supervision on encountered failures and preserves the deployment interface of the one-step policy.

Failure-state collection. After behavior-prior initialization, the Gaussian policy is rolled out using deployable observations o_t . For each predicted action chunk, a privileged collision model [21] checks scene and self-collision over the whole-body configurations along the horizon. A collision reported by either check triggers storage of the learner state, range observation, goal, and local action context. These samples form the learner buffer $\mathcal{D}_{\text{learner}}$ and identify the states that require geometric correction. Repeated rollout collection augments the buffer as the policy evolves, keeping post-training focused on its encountered failure distribution. During parameter updates, a mixed loader pairs learner-state samples with original samples from $\mathcal{D}_0$ at a 1:1 batch ratio.

Differentiable geometric supervision. Scene geometry and robot kinematics are available while constructing the post-training loss. As illustrated in Fig. 4, we represent the robot by collision spheres and evaluates every predicted waypoint in the action chunk. The local policy output is first transformed to the corresponding world-frame whole-body configuration:

$$q_\tau = T_{\text{lw}}(a_\tau), \quad \mathcal{S}(q_\tau) = \{(\mathbf{c}_{\tau j}(q_\tau), r_j)\}_{j=1}^{N_s}, \quad (9)$$

where T_{lw} is the local-to-world transformation and $\mathbf{c}_{\tau j}$ is the center of collision sphere j from forward kinematics.

Let $s_\mathcal{E}(\mathbf{c})$ be the scene signed distance, positive in free space, and let η be the collision activation margin. For each sphere, define $\rho_{\tau j} = r_j + \eta - s_\mathcal{E}(\mathbf{c}_{\tau j})$. The checker applies the piecewise-smooth penalty

$$\varphi_\eta(\rho) = \begin{cases} 0, & \rho \leq 0, \\ \rho^2 / (2\eta), & 0 < \rho \leq \eta, \\ \rho - \eta/2, & \rho > \eta, \end{cases} \quad (10)$$

The scene cost aggregates this penalty over the action horizon and robot spheres:

$$d_{\text{scene}} = \frac{1}{H N_s} \sum_{\tau=1}^H \sum_{j=1}^{N_s} \varphi_\eta(\rho_{\tau j}). \quad (11)$$

With the zero-margin limiting case $\varphi_0(\rho) = [\rho]_+$, we set $\eta = 0$ during post-training. Thus, the scene cost is zero in free space and at exact sphere-environment contact, and increases linearly with penetration.

For self-collision, let $\mathcal{P}$ contain the enabled sphere pairs and let $\tilde{r}_j$ include the configured sphere padding. We use the maximum positive overlap at each waypoint:

$$d_{\text{self}} = \frac{1}{2H} \sum_{\tau=1}^H \max_{(i,j) \in \mathcal{P}} [(\tilde{r}_i + \tilde{r}_j)^2 - \|\mathbf{c}_{\tau i} - \mathbf{c}_{\tau j}\|_2^2]_+. \quad (12)$$

We combine the collision penalties as:

$$\mathcal{L}_{\text{Priv.}} = \lambda_{\text{scene}} \mathbb{E}[d_{\text{scene}}] + \lambda_{\text{self}} \mathbb{E}[d_{\text{self}}], \quad (13)$$

The SDF query supplies gradients with respect to sphere centers, and the sphere-pair term has analytic gradients for the active pair. Automatic differentiation then connects the checker to the policy:

$$\nabla_{\theta} \mathcal{L}_{\text{Priv.}} = \frac{\partial \mathcal{L}_{\text{Priv.}}}{\partial \mathcal{S}} \frac{\partial \mathcal{S}}{\partial q} \frac{\partial q}{\partial \mathbf{A}_{\theta}} \frac{\partial \mathbf{A}_{\theta}}{\partial \theta}. \quad (14)$$

This gradient directly updates action chunks that produced collision-triggered states. In the reported configuration, $\lambda_{\text{scene}} = \lambda_{\text{self}} = 1$. The geometric term is evaluated on learner-state batches, whereas Eq. 8 is evaluated on the paired $\mathcal{D}_0$ batches.

Post-training objective. With the DDPM behavior-prior parameters ϕ frozen, post-training corrects geometric failures while bounding deviation from the initialized behavior. This objective can be written as the constrained problem

$$\begin{aligned} \min_{\theta} \quad & \mathbb{E}_{o \sim \mathcal{D}_{\text{learner}}} [\mathcal{L}_{\text{Priv.}}(\theta; o)] \\ \text{s.t.} \quad & \mathbb{E}_{o \sim \mathcal{D}_0} [\mathcal{L}_{\text{TR}}(\theta; o)] \leq \delta, \end{aligned} \quad (15)$$

where the first term corrects learner-induced geometric failures and the constraint defines a behavior trust region on the original observation distribution. In implementation, we optimize its penalty form on mixed batches:

$$\begin{aligned} \mathcal{L}_{\text{post}} = \mathbb{E}_{o \sim \mathcal{D}_0} [\mathcal{L}_{\text{TR}}(\theta; o)] \\ + \beta \mathbb{E}_{o \sim \mathcal{D}_{\text{learner}}} [\mathcal{L}_{\text{Priv.}}(\theta; o)]. \end{aligned} \quad (16)$$

Each mixed update estimates the two expectations from equally sized sub-batches. The first term anchors the policy to the pretrained behavior distribution, and the second applies corrective gradients at learner-induced collision states. The reported configuration uses $\beta = 1$. Learner states receive supervision directly from geometric distances, while the behavior term stabilizes updates when the correction buffer is small. Deployment retains the Gaussian policy and its deployable inputs; the privileged collision model remains a training-time component.

IV. EXPERIMENTS

This section evaluates GeoPT through three questions: (1) does GeoPT improve planning quality and efficiency over policy baselines; (2) does collision-triggered privileged post-training improve safety while preserving the behavior prior; and (3) does GeoPT generalize to unseen layouts in real-time deployment?

A. Experimental Setup

The expert-data construction and collision-triggered sample collection are described in Sections III-B and III-D. All methods receive the same first-person observation o_t and predict an action chunk of length $H = 50$. Each scene uses 20 shared initial robot poses and two trials per pose, yielding 40 trials per method.

Baselines. We compare the iterative DDPM behavior prior, Flow Matching, a one-step Gaussian policy trained by direct action regression (GeoPT-MSE), the one-step policy after behavior-prior initialization but before geometric post-training (GeoPT-TR), the complete method (GeoPT-Full),

Algorithm 1: GeoPT training procedure

Input: $\mathcal{D}_0$, DDPM behavior prior π_{ϕ} , Gaussian policy π_{θ} , collision checker $\mathcal{C}$, N_{post} , α , β
Output: Post-trained one-step policy π_{θ}

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// Train and freeze the behavior
prior.
1  $\phi \leftarrow \arg \min_{\phi} \mathcal{L}_{\text{DDPM}}(\phi)$ ; freeze( $\phi$ ) (Eq. 5)
// Initialize the Gaussian policy.
2  $\theta \leftarrow \arg \min_{\theta} \mathcal{L}_{\text{TR}}(\theta; \phi)$  (Eq. 8)
3  $\mathcal{D}_{\text{learner}} \leftarrow \emptyset$ 
// Define the collision trigger.
4  $\chi_{\mathcal{C}} \leftarrow \mathbb{I}[d_{\text{scene}} + d_{\text{self}} > 0]$  (Eqs. 11–12)
5 for  $i \leftarrow 1$  to  $N_{\text{post}}$  do
    // Collect rollout samples.
6  $\mathcal{R}_i \leftarrow \{(o_t, \mathbf{A}_t) : \mathbf{A}_t \sim \pi_{\theta}(\cdot | o_t)\}$ 
7  $\mathcal{F}_i \leftarrow \{(o_t, \mathbf{A}_t) \in \mathcal{R}_i : \chi_{\mathcal{C}}(\mathbf{A}_t, o_t) = 1\}$ 
8  $\mathcal{D}_{\text{learner}} \leftarrow \mathcal{D}_{\text{learner}} \cup \mathcal{F}_i$ 
    // Sample balanced batches.
9  $\mathcal{B}_0 \sim \mathcal{D}_0$ ,  $\mathcal{B}_{\ell} \sim \mathcal{D}_{\text{learner}}$ ,  $|\mathcal{B}_0| = |\mathcal{B}_{\ell}|$ 
    // Evaluate mixed loss.
10  $\hat{\mathcal{L}}_{\text{post}} \leftarrow \mathcal{L}_{\text{TR}} + \beta \mathcal{L}_{\text{Priv.}}$  (Eq. 16)
11  $\theta \leftarrow \theta - \alpha \nabla_{\theta} \hat{\mathcal{L}}_{\text{post}}$ 
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and M^2 Diffuser [3]. Flow Matching uses the same conditional encoder and DiT backbone as the DDPM behavior prior, changing only the generative training objective and sampling process. We implement M^2 Diffuser as a local planner with the same first-person observation and action-chunk output interface. It uses a conditional DDPM with gradient-based guidance for trajectory optimization in explicit scene representations. Its reverse process uses a geometric objective consistent with GeoPT’s privileged collision gradients.

Metrics. We report success rate (SR), collision rate, replanning steps, base path length, arm joint distance, final base error, and final joint error. SR is the percentage of trials that reach the commanded waypoint without collision. Replanning steps denotes the number of local replanning cycles executed within a trial; each cycle receives a new observation and produces an action chunk, with at most 30 cycles in the current benchmark. The position, yaw, and maximum joint-error thresholds are 0.1 m, 0.1 rad, and 0.1 rad, respectively. Collision rate is the fraction of local replanning cycles that trigger a scene or self-collision. Arm joint distance is the accumulated L1 displacement in configuration space. Base error is the Euclidean distance between the final and target base positions, reported in meters, while joint error is the maximum absolute error between the final and target arm-joint configurations, reported in radians. Final errors are averaged over collision-free trials with valid terminal-error measurements; the remaining continuous metrics are averaged over all trials.

Implementation details. All models use PyTorch and AdamW. Both the DDPM behavior-prior training and Gaussian initialization are run for 1024 epochs, with 50 diffusion

TABLE I
QUESTION 1: SIMULATION COMPARISON OF PLANNING QUALITY AND EFFICIENCY.

Scene split	Method	SR (%)	Coll. rate (%)	Base path (m)	Joint dist. (rad)	Base error (m)	Joint error (rad)	Replanning steps
Seen	DDPM behavior prior	25.0	19.9	3.05	3.73	0.022	0.003	3.53
	Flow Matching	45.0	12.9	4.05	4.92	0.041	0.015	4.28
	M^2 Diffuser	40.0	13.5	3.85	4.63	0.041	0.009	4.25
	GeoPT-MSE	55.0	15.6	4.84	7.02	0.293	0.019	4.80
	GeoPT-TR	52.5	14.8	4.72	6.27	0.052	0.020	4.78
	GeoPT-Full	57.5	7.1	4.79	6.63	0.044	0.016	4.88
Novel	DDPM behavior prior	17.5	24.8	3.22	3.98	0.035	0.006	3.70
	Flow Matching	37.5	15.8	4.24	5.14	0.052	0.019	4.45
	M^2 Diffuser	35.0	16.5	4.06	4.91	0.055	0.014	4.43
	GeoPT-MSE	45.0	17.1	5.06	7.35	0.318	0.025	5.03
	GeoPT-TR	40.0	16.4	4.94	6.55	0.067	0.026	5.00
	GeoPT-Full	50.0	9.9	5.05	6.91	0.061	0.022	5.10

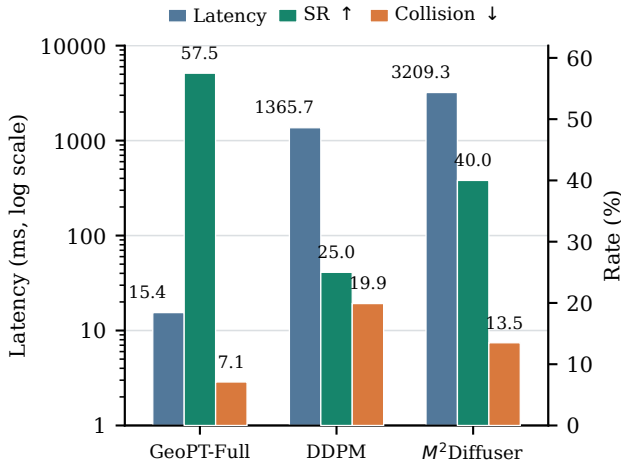


Fig. 5. **Latency-performance comparison.** Planning latency, success rate, and collision rate for GeoPT-Full, DDPM, and M^2 Diffuser.

steps and action chunk size 50. Privileged post-training runs for 64 outer iterations, with 32 epochs per iteration, using unit behavior and collision weights and a 1:1 mixture of original and learner-state batches.

B. Question 1: Does GeoPT Improve Planning Performance and Efficiency?

We evaluate the methods on the same set of seen layouts, using a shared observation, action-chunk interface, controller, and evaluation horizon for all policies. Table I compares the DDPM behavior prior, Flow Matching, GeoPT-MSE, GeoPT-TR, GeoPT-Full, and M^2 Diffuser using the metrics defined above. Fig. 5 summarizes the latency-performance trade-off. GeoPT-Full achieves the highest SR and lowest collision rate while retaining one-step inference. Relative to Flow Matching, it improves SR by 12.5 percentage points and reduces the collision rate by 5.8 points. The latency values use the end-to-end Orin deployment measurements reported in Table IV.

The DDPM teacher has lower SR because expert actions concentrate around a few high-quality modes rather than a broad multimodal distribution. Covering the full conditional distribution by stochastic diffusion introduces unnecessary variance that can accumulate over replanning, whereas Gaussian mean deployment avoids this issue. Although GeoPT-

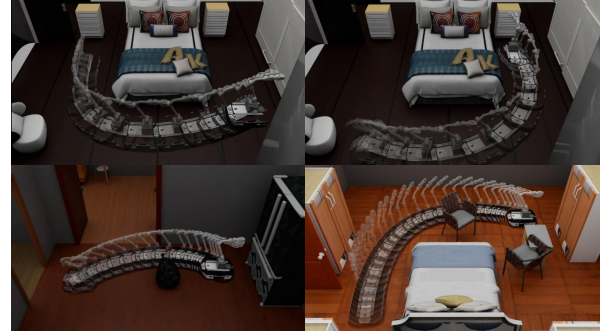


Fig. 6. **Visualization of planned trajectories.**

MSE attains slightly higher initial SR, exact regression tightly anchors the policy to individual labels. The distributional trust region instead preserves high-density behavior while allowing collision-guided updates toward higher-value, collision-free actions.

C. Question 2: Does Privileged Post-Training Improve Safety Without Losing Behavior?

Question 2 focuses on the causal contribution of collision-triggered post-training. We compare GeoPT-TR with GeoPT-Full and with controlled variants that remove learner-state collection or the behavior-preservation term. The evaluation uses the same rollout protocol and reports SR, collision rate, and the maximum collision cost over evaluation rollouts. We define collision cost as the sum of the scene and self-collision costs at each rollout step, then report the maximum summed value.

Table II reports these ablations under the same evaluation protocol. GeoPT-TR is the initialized one-step policy before geometric post-training. To isolate the value of collecting failure states, “GeoPT w/o learner-state collection” uses the same post-training schedule and privileged geometric objective as GeoPT-Full, but evaluates both $\mathcal{L}_{TR}$ and $\mathcal{L}_{Priv}$ on original-data states sampled from $\mathcal{D}_0$; no collision-triggered rollout state enters its updates. GeoPT-Full instead evaluates $\mathcal{L}_{Priv}$ on $\mathcal{D}_{learner}$ as defined in Eq. 16. This design distinguishes the effects of privileged geometry on expert-distribution states, targeted learner-state correction, and behavior preservation.

Fig. 7 tracks SR and collision rate throughout the two

TABLE II
QUESTION 2: ABLATION OF PRIVILEGED POST-TRAINING AND BEHAVIOR PRESERVATION.

Variant	Learner states	Priv. geometry	Behavior TR	SR (%)	Coll. rate (%)	Max. collision cost
GeoPT w/o learner-state collection	No	Yes	Yes	45.0	18.0	0.31
GeoPT w/o behavior preservation	Yes	Yes	No	0.0	31.6	0.42
GeoPT-Full	Yes	Yes	Yes	57.5	7.1	0.16

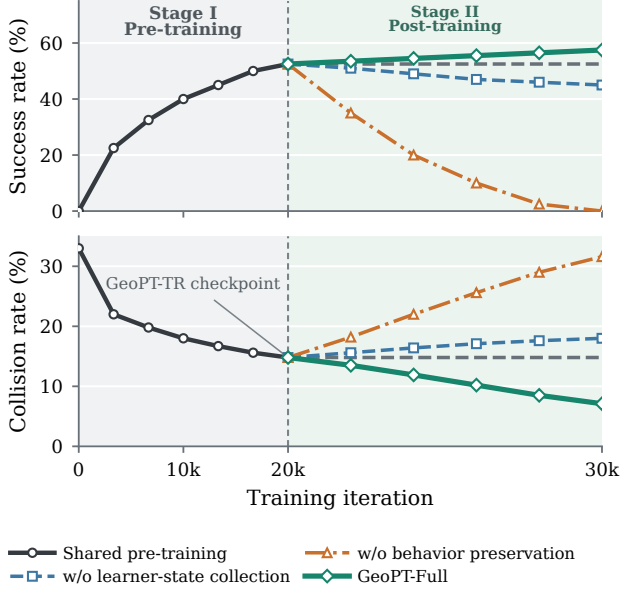


Fig. 7. **Two-stage training dynamics.** Success rate (top) and collision rate (bottom) across shared behavior-prior initialization and privileged post-training.

training stages. The variants share the behavior-prior initialization trajectory and branch from the GeoPT-TR checkpoint when privileged post-training begins. GeoPT-Full improves both metrics steadily, whereas training without learner-state collection overfits the original data: SR drops below the approximately 50% Gaussian-TR reference while collision rate rises. Training without behavior preservation rapidly collapses: SR falls to zero as collision rate increases, and goal-directed behavior is lost.

For the variant without behavior preservation, the privileged collision loss $\mathcal{L}_{\text{Priv}}$ in Eq. 13 decreases rapidly after geometric supervision is introduced, indicating effective correction of the sampled learner states. This local improvement does not yield stable closed-loop behavior: relative to the pre-trained policy, the action distribution drifts progressively during deployment. Although the corrected policy avoids the states already represented in its learner buffer, the shifted actions drive subsequent rollouts into regions not covered by either the original dataset or the finite post-training collection. The resulting compounding distribution shift substantially increases the closed-loop collision rate and cannot be compensated for within the finite collection rounds. This behavior explains why privileged correction requires the frozen behavior prior to constrain policy drift.

TABLE III

QUESTION 3: PHYSICAL-ROBOT DEPLOYMENT RESULTS OVER 20 TRIALS PER SCENE.

Scene	Replanning steps	Mean SR (%)	Trial coll. rate (%)
Seen scene 1	7.91	40.0	10.0
Seen scene 2	8.33	40.0	15.0
Novel scene 1	8.84	35.0	20.0
Novel scene 2	9.27	30.0	35.0

TABLE IV

QUESTION 3: LATENCY MEASURED ON THE JETSON AGX ORIN.

Stage / method	Mean latency (ms)	Frequency (Hz)	Measurement details
Raycasting construction	0.732	—	CPU raycasting: raw cloud $\rightarrow$ 30×180 image
Multimodal condition encoder	5.998	—	CNN + state/goal MLPs + fusion
One-step Gaussian policy	3.183	—	Gaussian action head
GeoPT-Full	15.440	64.8	One-step action generation
DDPM	1365.724	0.7	50-step denoising
M^2 Diffuser	3209.252	0.3	50-step denoising + geometric guidance

D. Question 3: Does GeoPT Generalize in Real Time?

We evaluate generalization on held-out layouts in Table I and deploy GeoPT on a 6-DoF Unitree Z1 manipulator mounted on an AgileX Tracer differential-drive chassis. FAST-LIO2 [22] estimates the robot state from a Livox MID-360 LiDAR, and the policy runs on an NVIDIA Jetson AGX Orin. Each trial executes a complete mobile-manipulation cycle: the robot first performs a goal-reaching task from the start pose to the target, with collisions recorded during this phase; a wrist-mounted camera then estimates the object pose for picking; finally, the robot places the grasped object at the specified pose. Fig. 8 shows this execution sequence.

We evaluate two seen and two novel scenes with 20 trials per scene. The reported SR is the arithmetic mean of the goal-reaching, pick, and place success rates. A trial is counted as a collision trial if at least one collision occurs during the goal-reaching phase, and the trial collision rate is its fraction among the 20 trials. Replanning steps also characterize this phase. Table III reports these physical-robot results. Table IV reports the Gaussian stage breakdown and end-to-end latency for GeoPT-Full, DDPM, and M^2 Diffuser on the Orin, averaged over 20 timed iterations after warm-up.

These results demonstrate that GeoPT generalizes to unseen layouts and supports real-time whole-body deployment on a physical robot.

V. CONCLUSION

We presented GeoPT, a privileged geometric post-training framework for one-step whole-body motion planning. GeoPT initializes a Gaussian planner from a DDPM behavior prior, collects collision-triggered learner states, and corrects them with privileged scene and self-collision supervision while constraining behavior drift. Privileged geometry is removed

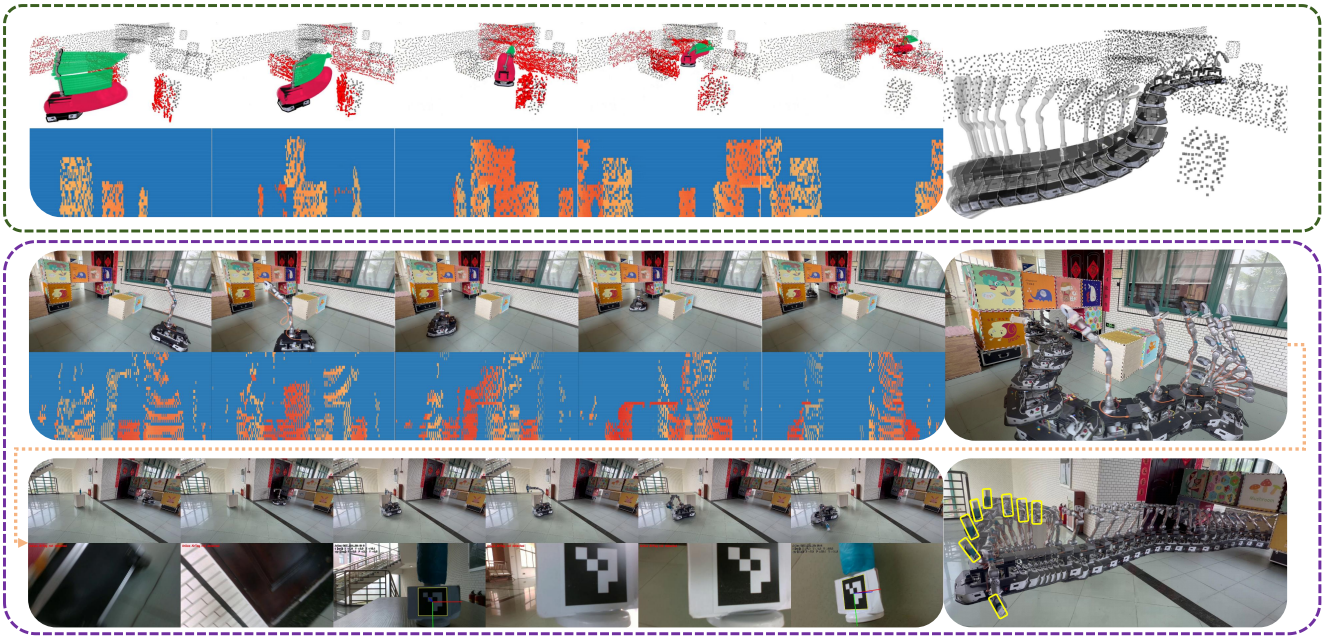


Fig. 8. **Task results visualization.** **Top:** Simulated goal-reaching task. **Bottom:** Complete real-world task cycle, including goal-reaching, wrist-camera-guided object picking, and placement at the specified pose.

after training, retaining single-pass deployment. The method provides no formal collision guarantee and depends on scene coverage and geometric-supervision fidelity. Future work will expand the scale and diversity of mobile-manipulation data to cover a broader range of environments, tasks, and whole-body interactions.

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